

K-Nearest Neighbor Algorithm

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Outline

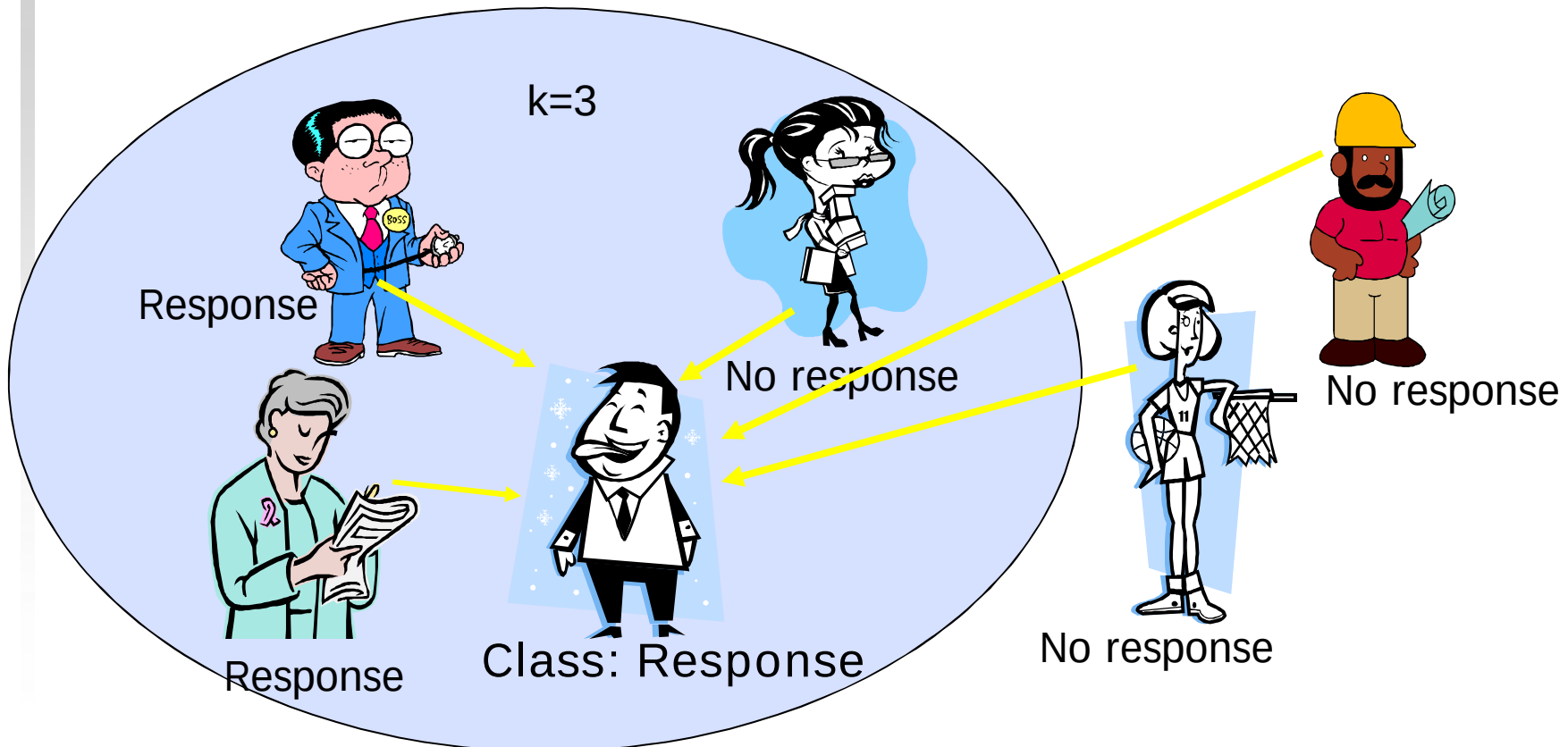
- ▶ KNN description with examples
- ▶ Distance measures
- ▶ Three issues and their solutions
- ▶ Pros and cons

K-Nearest Neighbor Algorithm

- ▶ A supervised learning method
 - ▶ A set of *labeled* training examples is given
 - ▶ Can do both classification and regression
- ▶ It does not learn a model (no learning or induction)
- ▶ To make a prediction on a new example, it
 - ▶ finds k most similar examples in the training data
 - ▶ use these examples' target values to figure out the prediction for the new example

K-Nearest Neighbor Classifier

- ▶ To determine the class of a new example E:
 - ▶ Calculate the distance between E and all examples in the training set
 - ▶ Select k nearest examples to E in the training set
 - ▶ Assign E to the most common class among its k -nearest neighbors

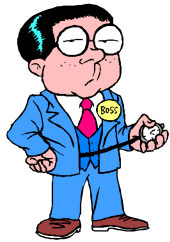


K-Nearest Neighbor Learning

- ▶ **Instance/Memory-Based/Lazy Learning**
 - ▶ No model is built: Store all training examples
 - ▶ Any processing is delayed until a new instance must be classified.
- ▶ **Lazy vs. Eager Learning**
 - ▶ **Lazy:**
 - ▶ No training phase or model. Use training data during prediction
 - ▶ **Eager:**
 - ▶ Construct a generalization model from training data and use it for prediction
- ▶ **Inductive Assumption of KNN**
 - ▶ Similar inputs map to similar outputs
 - ▶ If not true, learning is impossible
 - ▶ If true, learning reduces to computing similarity/distance

Similarity or Distance Measures

- ▶ Assume that each example is represented with a set of numerical attributes



John:
Age=35
Income=95K
No. of credit cards=3



Rachel:
Age=41
Income=215K
No. of credit cards=2

- ▶ “Closeness” can be measured by the ***Euclidean distance*** between two examples $X = (x_1, x_2, x_3, \dots, x_n)$ and $Y = (y_1, y_2, y_3, \dots, y_n)$:

$$D(X, Y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

- ▶ Distance (John, Rachel):

$$\sqrt{(35-41)^2 + (95K-215K)^2 + (3-2)^2}$$

Example : 3-Nearest Neighbors

Customer	Age	Income	No. credit cards	Response
John 	35	35K	3	No
Rachel 	22	50K	2	Yes
Hannah 	63	200K	1	No
Tom 	59	170K	1	No
Nellie 	25	40K	4	Yes
David 	37	50K	2	?

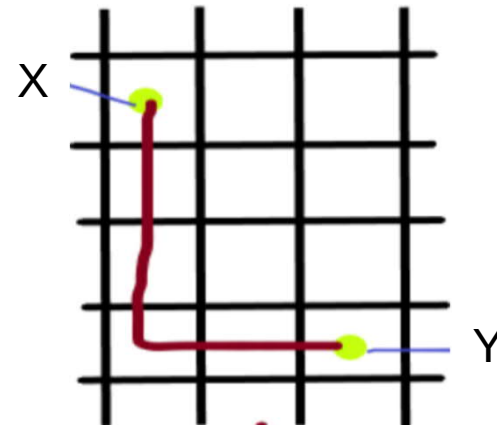
Example (3-NN)

Customer	Age	Income (K)	No. cards	Response	Distance from David
John 	35	35	3	No	$\text{sqrt} [(35-37)^2 + (35-50)^2 + (3-2)^2] = 15.16$
Rachel 	22	50	2	Yes	$\text{sqrt} [(22-37)^2 + (50-50)^2 + (2-2)^2] = 15$
Hannah 	63	200	1	No	$\text{sqrt} [(63-37)^2 + (200-50)^2 + (1-2)^2] = 152.23$
Tom 	59	170	1	No	$\text{sqrt} [(59-37)^2 + (170-50)^2 + (1-2)^2] = 122$
Nellie 	25	40	4	Yes	$\text{sqrt} [(25-37)^2 + (40-50)^2 + (4-2)^2] = 15.74$
David 	37	50	2	Yes	

Other Distance Measures (Numeric Attributes)

- ▶ **Manhattan distance:** Given two examples $X = (x_1, x_2, x_3, \dots, x_n)$ and $Y = (y_1, y_2, y_3, \dots, y_n)$:

$$D(\mathbf{X}, \mathbf{Y}) = \sum_{i=1}^n |x_i - y_i|$$

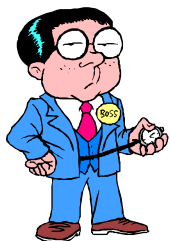


- ▶ **Cosine similarity:**

$$\text{Similarity}(\mathbf{X}, \mathbf{Y}) = \cos(\theta) = \frac{\mathbf{X} \cdot \mathbf{Y}}{\|\mathbf{X}\| \|\mathbf{Y}\|}$$

where θ is the angle between $\mathbf{X}$ and $\mathbf{Y}$

A Problem and its Solution



John:

Age=35

Income=35K

No. of credit cards=3



Rachel:

Age=22

Income=50K

No. of credit cards=2

$$\text{Distance (John, Rachel)} = \sqrt{(35-22)^2 + (35000-50000)^2 + (3-2)^2}$$

- ▶ Distance between examples could be dominated by some attributes with relatively large numbers (e.g., income in our example).
- ▶ Important to normalize features (e.g., map numbers to numbers between 0-1)
 - ▶ If Highest income = 200K and Lowest income=0, use Min-Max normalization:
 - John's income is normalized to 35/200
 - Rachel's income is normalized to 50/200

KNN with Normalization of Variables

Customer	Age	Income (K)	No. cards	Response
John 	$55/63 = 0.55$	$35/200 = 0.175$	$\frac{3}{4} = 0.75$	No
Rachel 	$22/63 = 0.34$	$50/200 = 0.25$	$2/4 = 0.5$	Yes
Hannah 	$63/63 = 1$	$200/200 = 1$	$\frac{1}{4} = 0.25$	No
Tom 	$59/63 = 0.93$	$170/200 = 0.85$	$\frac{1}{4} = 0.25$	No
Nellie 	$25/63 = 0.39$	$40/200 = 0.2$	$4/4 = 1$	Yes
David 	$37/63 = 0.58$	$50/200 = 0.25$	$2/4 = 0.5$	Yes

Another Problem and its Solution

- ▶ Distance works naturally with numerical attributes
- ▶ What if we have nominal attributes?

Example: Married

Customer	Married	Income (K)	No. cards	Response
John	Yes	35	3	No
Rachel	No	50	2	Yes
Hannah	No	200	1	No
Tom	Yes	170	1	No
Nellie	No	40	4	Yes
David	Yes	50	2	

K-NN with Nominal Attributes

▶ Method 1:

- ▶ Convert a nominal attribute to a numerical attribute, for example:
 - ▶ $\text{yes} \Rightarrow 1$ and $\text{no} \Rightarrow 0$
 - ▶ $\text{blue} \Rightarrow 1$, $\text{yellow} \Rightarrow 2$, $\text{red} \Rightarrow 3$, etc.
- ▶ Problem?

▶ Method 2:

- ▶ Convert a nominal attribute to n binary attributes, where n is the number of values of the nominal attribute
 - ▶ For example, if the domain of attribute A is $\{a_1, a_2, a_3\}$, we convert it into 3 attributes to represent A .
 - ▶ If $A=a_1$, the values of the three attribute are 1, 0, 0.
 - ▶ If $A=a_2$, the values of the three attribute are 0, 1, 0.
 - ▶ If $A=a_3$, the values of the three attribute are 0, 0, 1.

*One-hot
encoding*

K-NN with Nominal Attributes

- ▶ Method 3:

- ▶ Do not convert, but change the distance calculation:

$$\text{Distance } (x, y) = \sum_{i=1}^m \text{dist}(x_i, y_i)$$

where m is the number of attributes, and

$$\text{dist}(x_i, y_i) = \begin{cases} 0 & \text{if } x_i \text{ and } y_i \text{ are nominal and } x_i = y_i \\ 1 & \text{if } x_i \text{ and } y_i \text{ are nominal and } x_i \neq y_i \\ | \text{norm}(x_i) - \text{norm}(y_i) | & \text{if } x_i \text{ and } y_i \text{ are continuous} \end{cases}$$

where $|\text{norm}(x)|$ is a normalized value of x .

Third Problem and its Solution

- ▶ Curse of dimensionality

- ▶ As the number of attributes increases, distance between examples becomes larger and more uniform.
- ▶ If the number of relevant attributes is fixed, increasing the number of less relevant attributes may make distance less reliable

$$D(X, Y) = \sqrt{\sum_{i=1}^{relevant} (x_i - y_i)^2 + \sum_{j=1}^{irrelevant} (x_j - y_j)^2}$$

- ▶ Solutions

- ▶ Feature selection
- ▶ Weighted distance with feature weights

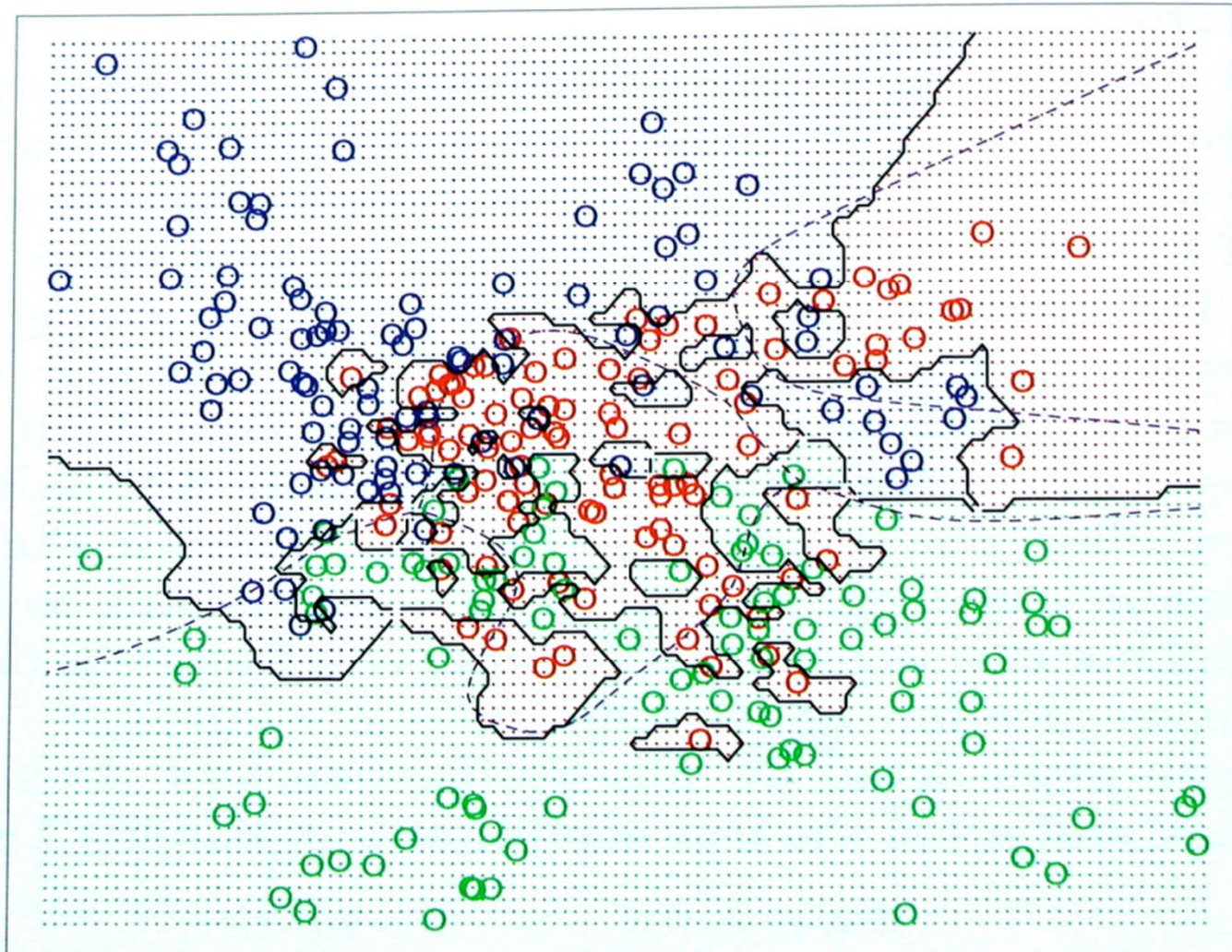
$$D(X, Y) = \sqrt{\sum_{i=1}^m w_i (x_i - y_i)^2}$$

How to Choose k

- ▶ Small k :
 - ▶ Sensitive to noise (particularly class noise)
 - ▶ Captures fine structure of space better
 - ▶ May be necessary with small training sets
- ▶ Large k :
 - ▶ Less sensitive to noise
 - ▶ Better class probability estimates for the example to be classified
 - ▶ Large training sets allow larger values of k
- ▶ Can use cross-validation on the training data to choose k

Class Boundaries

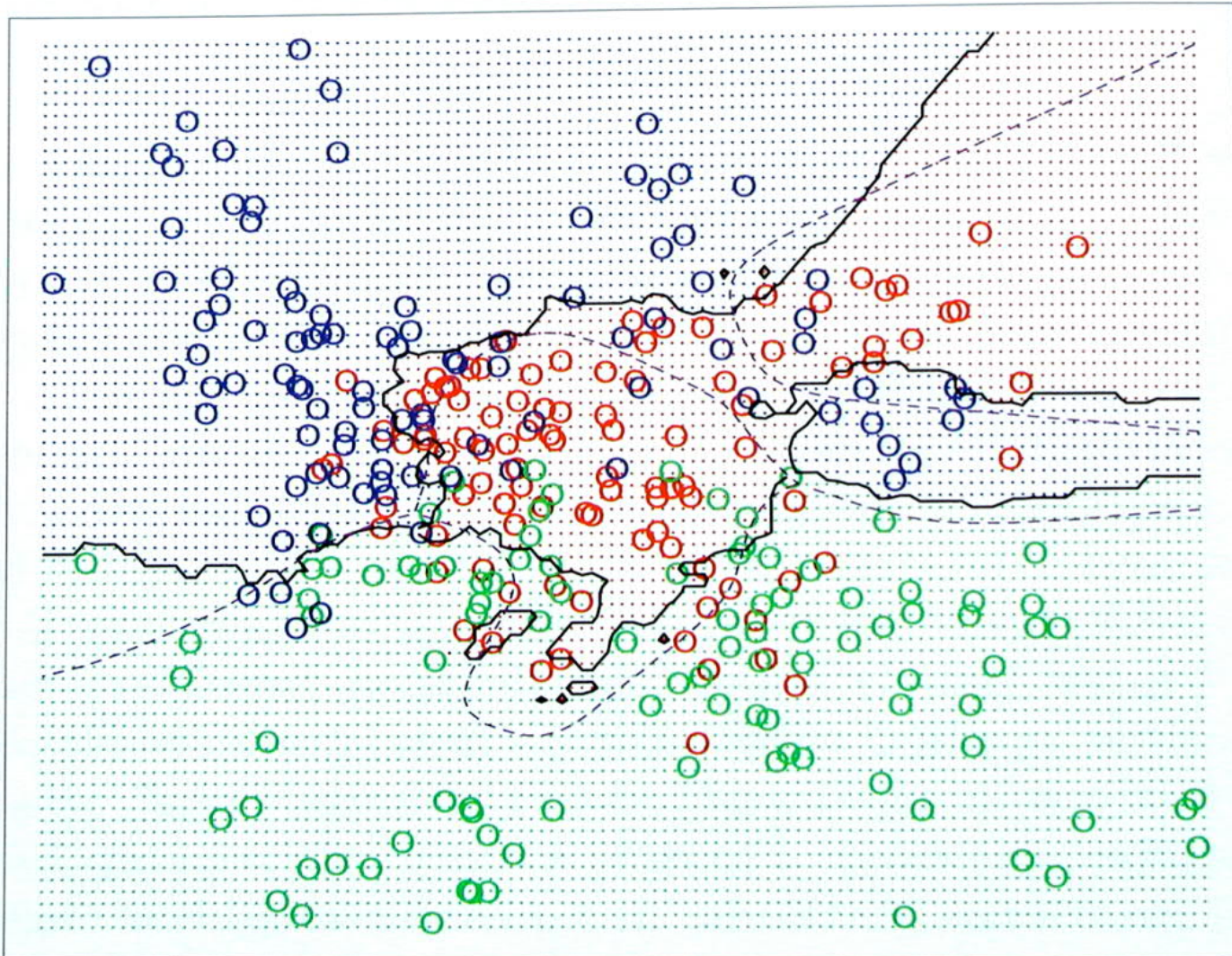
1-Nearest Neighbor



From Hastie, Tibshirani, Friedman 2001 p418

Class Boundaries

15-Nearest Neighbors



From Hastie, Tibshirani, Friedman 2001 p418

Strengths and Weaknesses

▶ Strengths:

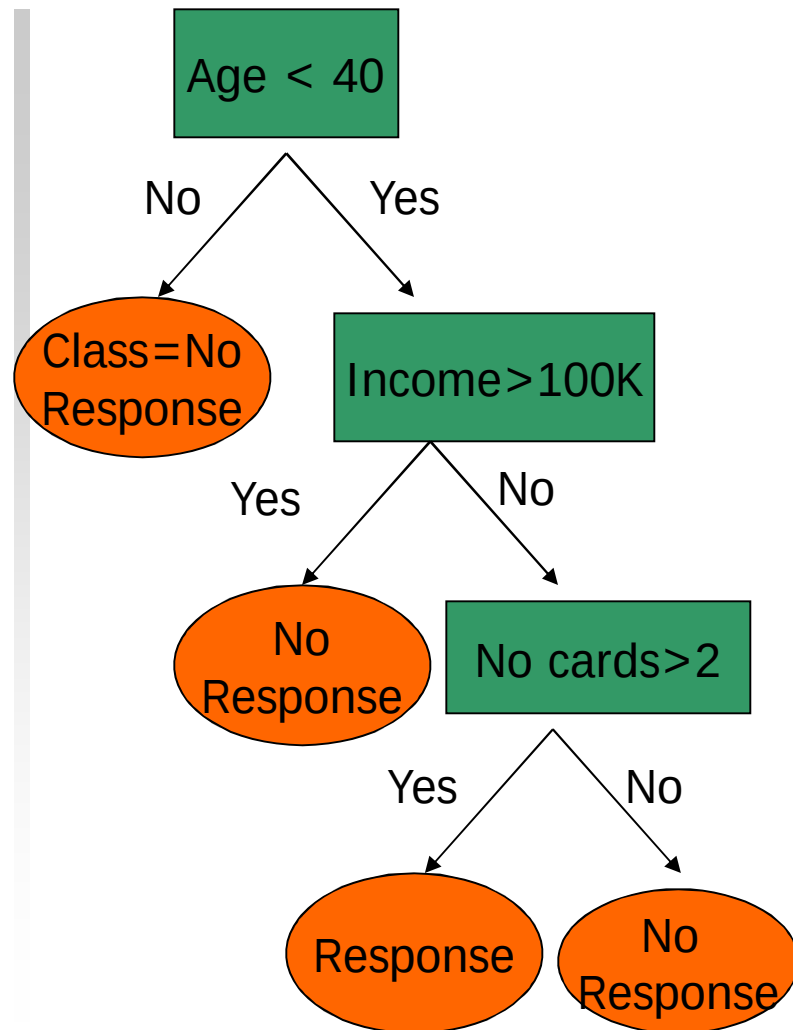
- ▶ Simple to implement and use
- ▶ Comprehensible – easy to explain prediction
- ▶ Robust to noisy data by averaging k-nearest neighbors.
- ▶ Can learn complex target functions
- ▶ Can be used for both classification and regression

▶ Weaknesses:

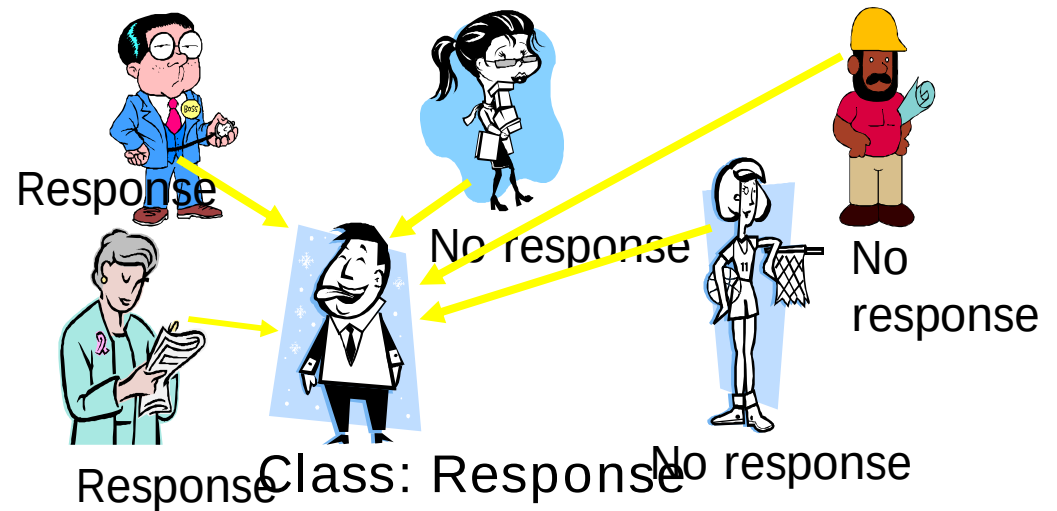
- ▶ Need a lot of space to store all examples.
- ▶ Takes more time to classify a new example than with a model
 - ▶ need to calculate and compare distance from new example to all other examples.

Decision Tree vs K-Nearest Neighbor Classifier

Classification Tree Model



K-Nearest Neighbors



Summary

- ▶ KNN description with examples
- ▶ Distance measures
- ▶ Three issues and their solutions
- ▶ Pros and cons

Summary

- ▶ KNN description with examples
- ▶ A commonly used distance measure
- ▶ Three issues and their solutions
- ▶ Pros and cons